

Stimulus in the Classroom, Signals in the Economy¹

Elana Wong²

24 April 2026

Abstract

This paper examines how American Recovery and Reinvestment Act (ARRA) instructional funds were allocated across states and whether these allocations influenced economic conditions. Using demographic, fiscal, and labor market data, estimations using OLS models aid to assess the determinants of ARRA funding and its relationship to changes in construction employment, unemployment, and median income. Results show that pre-ARRA socioeconomic characteristics strongly predict funding levels, while the economic impacts of ARRA vary across outcomes and states. These findings highlight both the targeting patterns and the heterogeneous effects of federal stimulus spending.

¹ I would like to thank Dr. Evans, for being a constant support system throughout my undergraduate and graduate career and Ryan Meyer for peer feedback and support.

² elanawong@gmail.com, (803) 448 – 4568, Clemson University, M.S. Economic Analytics

1.0 Introduction

The American Recovery and Reinvestment Act (ARRA) represented one of the largest federal stimulus efforts in U.S. history, directing resources to state and local governments post the Great Recession. A central component of this spending targeted public education systems, with the goal of stabilizing state education in attempts to support economic recovery. Yet despite the scale of these investments, relatively little is known about how instructional ARRA funds were distributed across states or whether these allocations generated measurable as well as statistically significant economic effects. This paper examines the determinants of ARRA instructional funding and evaluates its relationship to changes in construction employment, unemployment, and median household income. Using state-level data, a series of OLS models to assessed both the targeting of federal stimulus dollars and the local economic impacts that resulted. The results contribute to ongoing debates about the effectiveness of countercyclical fiscal policy and the channels through which education-related stimulus spending influences broader economic conditions.

2.0 Background and Literature Review

The American Recovery and Reinvestment Act provided stimulus funding for the United States Department of Education. These funds were aimed to deliver emergency education funding, especially in response to the Great Recession of 2009 but also used to spur innovation and reform at various levels of the educational system in the U.S. The ARRA funding varied largely across states from \$1,063 to \$3,632 per pupil. States with the largest budget shortfalls and states with higher achievement received more than states with smaller budget shortfalls and lower student achievement. The higher the poverty rate the higher the amount of funding a school district received from the ARRA. At the school and district level, progress was uneven and varied. The funds were allocated and provided to schools in 2010. The research question sought to be answered now is what exactly contributed to the determination of ARRA funding allocations, in terms of economic outcomes and then on the flipside, what did states who received higher levels of funding do to maximize the economic growth outcomes? Is it possible to see the economic outcomes from the ARRA spending in various growth factors?

2.1 Trabelsi

Trabelsi (2018) asks whether public education expenditure promotes economic growth and whether this relationship depends on institutional quality. Using cross-country panel data for 109 countries from 1980 to 2013, the study employs OLS and nonlinear structural threshold regression models, with governance indicators serving as the threshold variable. The study builds on the idea that the apparent inefficacy of public education expenditure is often rooted in government inefficiency, and Trabelsi integrates the two concepts of the direct link of public spending and economic growth with the acknowledgement that institutional quality is a driver of economic performance. The overarching thesis statement is that good governance increases the efficiency of public education investment, which in turn spurs growth, whereas weak governance leads to resource misallocation and negative outcomes. To understand and try to create a summation of institutional quality, the study employs five indicators from the World Governance Indicators (WGI) database: Political Stability, Government Effectiveness, Regulatory Quality, Rule of Law, and Control of Corruption. The thresholds for each of these institutional quality calculators were determined using Structural Threshold Regression or STR models which was critical to use because it accounts for the endogeneity of the threshold variables to avoid biased and inconsistent estimated parameters. The various thresholds were obtained in Table 3 for all five governance indicators. The results of the nonlinear model estimation show that education spending has a positive and statistically significant effect on growth only when a measure of governance exceeds a critical threshold, while below this threshold the effect becomes negative, implying that returns to spending are conditional and limited in weaker institutional environments. Because the analysis relies on observational cross-country data, the findings are more correlational rather than causal. A key strength of this study is its use of threshold modeling, which captures nonlinear relationships often missed in standard regressions. However, its reliance on cross-country data introduces substantial heterogeneity and potential omitted variable bias. Compared to later studies that use more localized or time-series approaches, this paper provides broad external validity but weaker internal validity. The overarching conclusion is that simply increasing resources for education is insufficient to guarantee economic development, but instead institutional reform remains a critical component of economic success. This study is directly related to this thesis in demonstrating that education spending alone may not contribute to positive economic growth unless complementary institutional conditions are met, emphasizing the importance of context.

2.2 Deskins

Deskins et al. (2009) provides examination of how different types of education expenditures influence U.S. state economic performance. This study seeks to clarify the relationship through two major innovations, disaggregating K - 12 from higher education spending and trying to account for cross-state spillover effects. The argument presented is that most research aggregates all education spending into a single category dismissing the differing impacts between primary and postsecondary education investments. They also address spatial correlation, where spending in one state may influence the economic growth of neighboring states through a movement of educated workers across state lines. Using panel data from 1992 to 2002, the authors apply fixed effects and instrumental variable regressions on 48 contiguous U.S. states. The researchers are cautious about controlling all elements of the state budgets such as federal transfers, noneducation spending and budget surpluses in an attempt to isolate the effect of education spending to specifically finance own - source revenues such as taxes. Using own-source revenues and allowing it to be the variable that changes, the impact of spending was isolated to the funds from the state's own taxpayers. The outcome variable of economic growth is measured using two distinct variables such as nongovernmental gross state product (GSP) and total private sector employment. The conclusions found a statistically significant negative effect on both GSP and employment growth indicating that an increase in per capita higher education spending is associated with a decrease in the state's GSP growth rate. In terms of K-12 spending, the results do not find a statistically significant relationship between primary and secondary education spending and economic growth. This result fails to support the hypothesis however it is considered that there may be a concern that the taxes used to fund education is offsetting the economic growth measurement. In terms of spillover effects, there are no statistically significant results supporting that claim. An insight of the study is that combined spending creates an overall negative effect, but the result is driven by the negative impact of higher education expenditures.

2.3 Clarke et al.

Clarke et al. (2015) further narrow the focus by asking whether education spending and workforce quality affect economic growth at the county level. Clarke et al. examines whether economic development at the county level is driven more by public education expenditures or the

educational attainment of the resident workforce. What this means is that simply increasing school funding is not a guaranteed path to growth but the area's ability to retain a high-quality workforce are primary determinants of positive economic growth outcomes. The goal of this study is to determine what is the truth from past studies that resulted in conflicting findings in terms of the link between education spending and economic growth. Some studies found that there is a positive relationship while others hypothesized that high taxes or inefficient administrative spending erase the economic benefits from the result. Clarke et al. hypothesizes that the workforce quality, measuring the proportion of residents with specific degrees, is a more accurate determinant of growth in contrast to school expenditures. Using pooled cross-sectional data from Georgia counties between 2006 and 2008, a specific benchmark before a major downturn in the US economy, the study applies OLS regression models to estimate the effects of K-12 spending and educational attainment on job growth and output. The study utilizes two primary dependent variables, percentage change in total jobs and percentage change in economic output, using specifically the biotechnology industry as a sector requiring a highly educated workforce, that was also a key growth industry. The analysis focuses on K - 12 school spending per pupil, and a proportion of the population holding either a high school degree or at least a bachelor's degree. The researchers use OLS models to control socio-economic factors such as poverty rates, unemployment and undeveloped land. The results show that K- 12 education spending itself is not statistically significant when determining job growth or economic output, while higher education attainment is positively associated with job growth, suggesting that workforce quality, rather than spending, drives economic outcomes, with meaningful magnitudes for college-level attainment. Increasing the college educated population by 10 percent is associated with a 1 percent increase in total employment growth and 1.7 percent increase in biotechnology jobs. High school graduates, according to this study, are negatively correlated to job growth. This indicates that high school graduates have fallen below the minimum requirements for specialized industries. The study is not causal, as it relies on observational data during the Great Recession, which may bias results. A strength of this paper is its micro-level focus, allowing for more detailed analysis of labor market dynamics. However, its short time frame and exposure to macroeconomic shocks weaken its conclusions. Like Trabelsi with the threshold hypothesis, there is likely another threshold about educational attainment that shifts the correlation from negative to positive, however that is likely crossing the threshold from working

on a college degree to actual completion. The conclusion of this study is that it depends more on the workforce qualities, not the expenditures put into the workforce, that matters for economic development. It is argued that K-12 spending is a necessary means to an end, but it's likely that the end will not benefit a local economy. The suggestion for the future is that governments focus on retaining highly educated individuals within a state which education is invested in, in order to reap the economic outcomes of having higher educated residents. This paper contributes to the thesis by suggesting that education spending may influence unemployment through human capital accumulation, which makes direct effects harder to detect.

2.4 Pirim et al.

More directly aligned with unemployment outcomes, Pirim et al. (2014) investigates whether investments in human capital, specifically through education and health expenditures, serve as effective tools for reducing unemployment in the United States. The researchers further divulge into the relationship between public spending and labor market outcomes, something that has historically produced mixed results. The study uses human capital theory, hypothesizing that higher quality human capital leads to higher performance and improved employment outcomes. Pirim et al. also utilizes arguments of positive externalities, potential neighborhood effects and governance quality, similar thinking to Trabelsi's theory. Using a 25-year panel of all states, the study applies OLS regressions with multiple specifications, including lagged variables and fixed effects. The control variables included fiscal effort, graduation rates, gross state product (GSP) per capita, welfare spending, political party affiliation and unionization status. The findings show that per pupil education spending is initially positively associated with unemployment. Health expenditures were also found to significantly reduce unemployment as well. Graduation rates showed a strong negative correlation with unemployment in simple ordinary least squares regression models; the effect became insignificant in fixed effects models suggesting that there are unaccounted for variables that likely impact the state-specific models. The study found that stronger unionization is positively related to unemployment in fixed-effects models, which authors hypothesize can be attributed to labor market rigidity. The results also suggested that a healthier and better educated workforce, in terms of quality education, is more flexible and resilient, which allows people to adjust better in times of economic downturn. In conclusion, Pirim et al. concluded that the most effective method for states to reduce unemployment in the

long term is to invest in the quality of human capital through health and education. Simply put, short-term outcomes can be misleading due to the nature of education being a long-term investment requiring patience as an input if tangible results are being sought out.

2.4 Mehmetaj and Xhindi

Mehmetaj and Xhindi (2022) build on the hypothesis of education and unemployment interaction by explicitly asking whether education spending has a causal effect on unemployment in the long run. The primary goal of the study examines whether the share of funds allocated to public education affects youth unemployment rates. The researchers primarily studied two models, total public education on economic growth and higher education spending on economic growth. The theoretical framework is the theory that education is a crucial component of economic progress and development of human capital is critical. This study combines the ideas from past studies of examining K -12 as well as secondary education spending outcomes. A case study done in Albania using time series data, employs a vector error correction (VEC) model to distinguish between short-run and long-run dynamics. The long-term impact of spending had strong causality with a 1 percent increase in total public spending associated with a 10.81 percent decrease in youth unemployment rate. This variable is different compared to past studies as it focuses more on K-12 spending and then youth unemployment rather than K-12 spending and young adult outcomes on unemployment. The higher education effect found that a 1 percent increase in higher education spending leads to a 5.85 percent decrease in unemployment rate. In terms of short-term dynamics, real economic growth rate shows causality with youth unemployment, but education expenditures do not. The data shows that the most effective long-term strategy for reducing youth unemployment is sustained investment in human capital.

2.4 Hanushek and Woessmann

Hanushek and Woessmann (2012) shift the focus from spending to outcomes, asking whether cognitive skills better explain economic growth than education inputs. The primary goal of the study sought to determine whether a causal relationship exists between cognitive skills and long-term economic growth instead of studying years of schooling. Using direct measures of student achievement, the researchers aim to address regions in the world where high school

achievement did not lead to expected economic gains. Using cross-country panel data of 50 countries between 1960 and 2000, the analysis utilized instrumental variables, difference in difference and longitudinal analysis. The instrumental variables were features of school systems like private school competition and teacher quality to identify variations regarding school characteristics. The difference in difference analysis was a comparison of the earnings of immigrants schooled in their home countries vs. schooled in the United States, isolating the impact of home country school quality. The results argue a powerful linked relationship between skills and growth. Variations in cognitive skills, in regard to test scores, completely describe regional growth differences, with the conclusion that one standard deviation increase in test scores is associated with a two percentage point higher annual growth rate in GDP per capita. Controlling for test scores, years of schooling become statistically insignificant in predicting growth. Showing that simply increasing school attendance without ensuring students learn is unlikely to produce economic returns. The researchers also find that both the share of high-performing students and the share of students reaching basic literacy are independently and significantly related to growth. High level skills are particularly important for developing countries to facilitate the imitation and adoption of new technologies. In conclusion, school quality is the primary driver of long run economic success, and they argue that there should be a shift to focus on institutional reforms instead of downright funding.

Utilizing the arguments from Clarke et al. about the biotechnology industry being used as a proxy for economic growth in an area, construction employment will be utilized instead because employment in the construction industry does not often require a college degree and therefore concerns about higher education standards can be discarded. Clarke et al. also takes socioeconomic outcomes into consideration, a concept repeated in this analysis. Since ARRA funding focused on K-12 students, it makes sense to choose an economic outcome that would rely more on K-12 quality schooling instead of a sector that requires higher educational attainment above twelfth grade. Similarly to Pirim et al., per pupil spending will be utilized in this research to further understand how schools are efficiently or inefficiently utilizing funding allocations. In contrast to the previous studies, an analysis calculating economic outcomes that were considered in terms of allocating ARRA funding will be run just to overview what economic outcomes contributed more strongly to various states receiving certain levels of funding.

3 Data Presentation

Most national datasets that provide the most comprehensive data as well as multiple states and years while being the most accurate are the governmental websites and federal databases. The National Center for Educational Statistics, US Census, Bureau of Labor Statistics (BLS), as well as the Bureau of Economic Analysis (BEA), are all commonly used throughout the study of the intersection of education and economics. While searching for the appropriate datasets for this analysis, the Federal Reserve of Economic Data provided insight, particularly with macroeconomic data such as construction employment, median household income and unemployment rates. All of the education data was collected from the National Center for Educational Statistics, through the various economic analysis charts and publications located on the NCES website.

3.1 Construction Employment

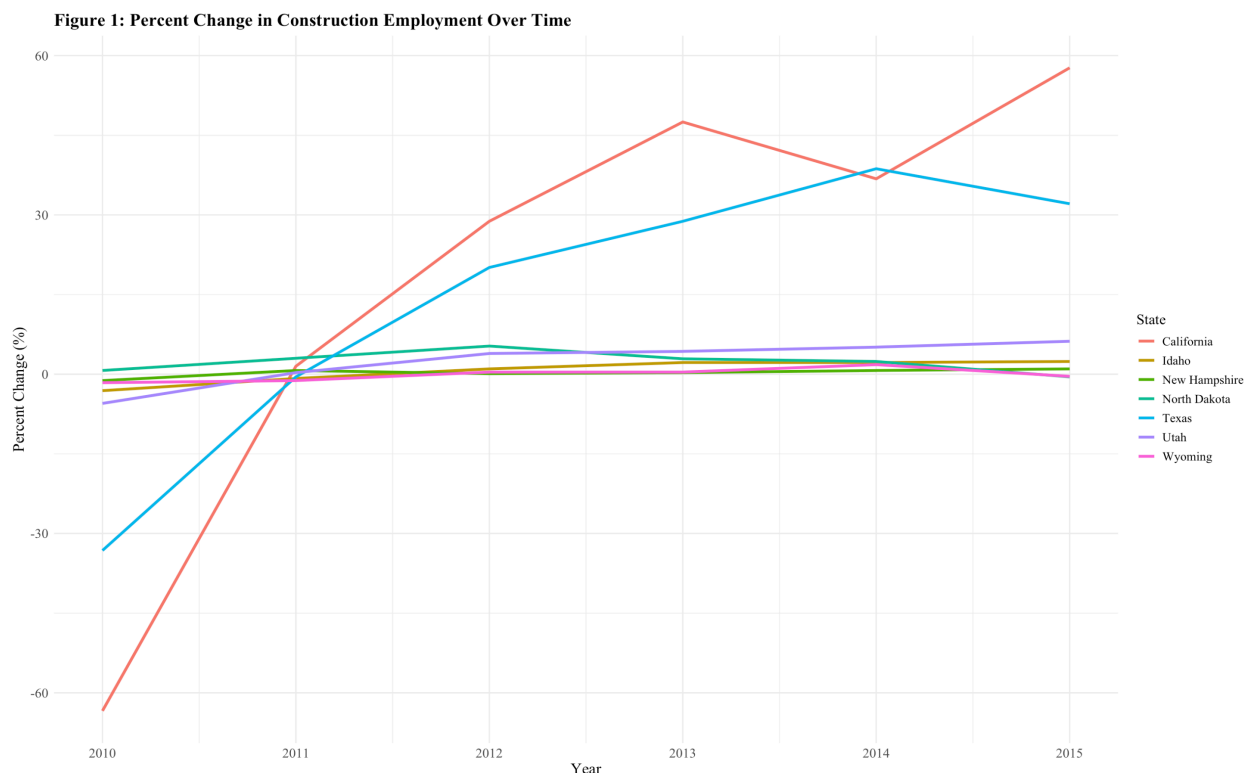
Table 1: Construction Employment Summary for Selected States

State	2005 Value	2015 Value	Total Percentage Change	Mean Annual Percentage Change
California	905.5	732.4	-19.1	-20.7
Idaho	45.0	38.2	-15.1	-13.4
New Hampshire	29.6	24.3	-17.9	-16.6
North Dakota	17.5	34.7	98.3	43.3
Texas	567.2	683.8	20.6	8.2
Utah	81.5	84.8	4.0	-2.1
Wyoming	20.7	23.3	12.6	12.9

The selected states above were selected because all of them received vastly different amounts of funding from the ARRA education funding. According to NCES ARRA data, Texas, California, and Utah experienced close to median change in funding (~6.7%). New Hampshire and Wyoming had two of the lowest percent changes (~0.5%). North Dakota and Idaho are the highest changes at 17.5% and 14.4%, respectively.

A macroeconomic indicator of economic growth within a state, construction employment changes over time can indicate growth in the economy via the increases in new infrastructure occurring across the state over time. The US Census provides information about most of the states with a general topic of “Construction Employees in ‘State’”, however this is not the case for all states. The construction sector comprises primarily the construction of buildings or engineering projects. Establishments primarily engaged in the preparation of sites for new

construction and establishments primarily engaged in subdividing land for sale as building sites also are included in this sector. There are three different subsectors underneath the sector of construction, construction of buildings, heavy and civil engineering construction and specialty trade contractors. When data was being collected, Delaware and Hawaii did not have the entire sector of construction listed on the Federal Reserve of Economic Data website, so a proxy was used instead. The graph below shows the significant changes in construction employment across specific states. These states were chosen because they are at the maximum, median and minimum for increases in instruction spending due to ARRA.

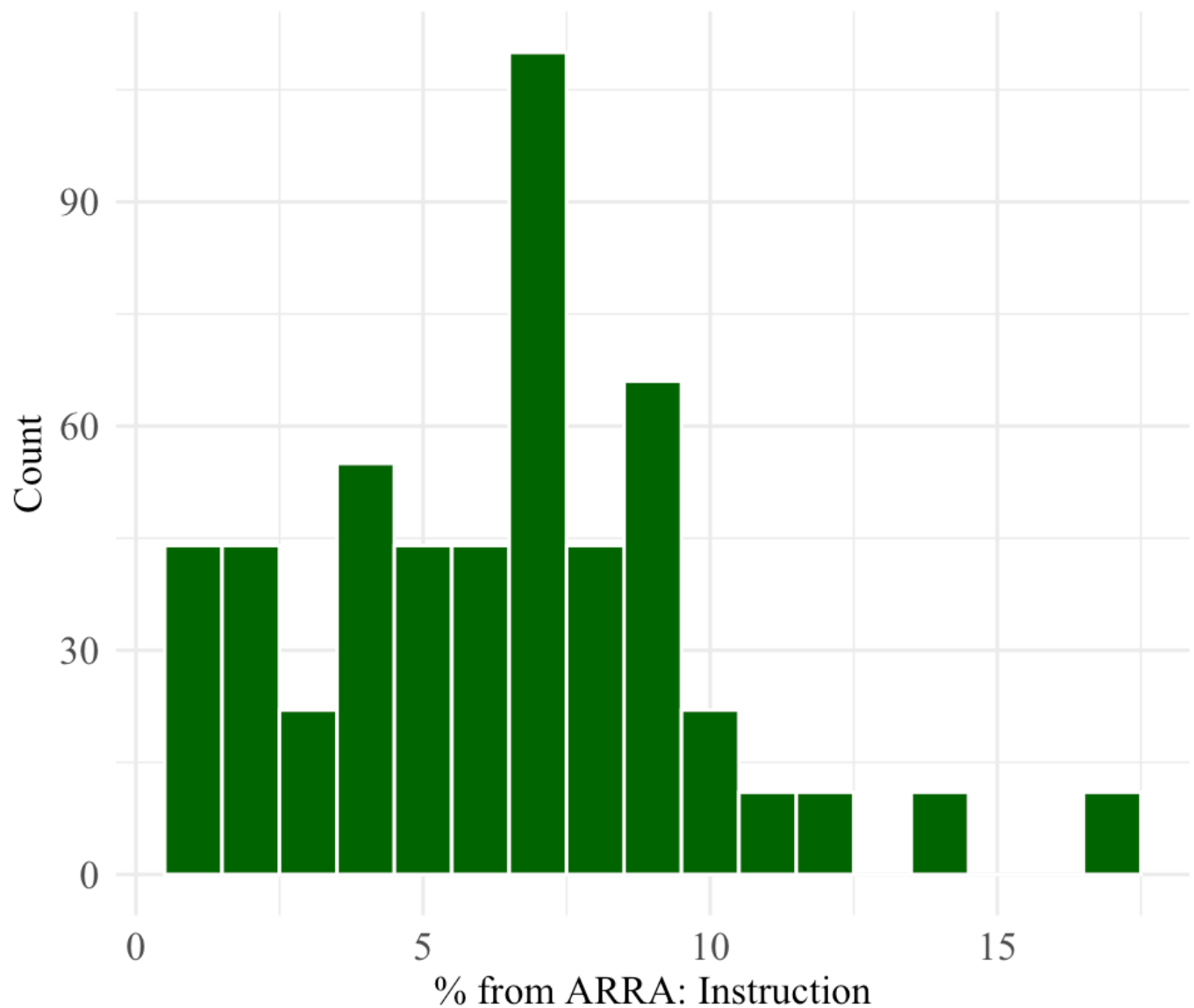


When data was being collected, Delaware and Hawaii did not have the entire sector of construction listed on the Federal Reserve of Economic Data website, so a proxy was used instead. Since construction employment right now is primarily being used as an indicator variable for economic growth, the subsector of construction of buildings was used. It is appropriate in the manner that new buildings tend to be a signal that economic development is happening in a state.

3.2 Number of Title 1 Schools

All schools included in the dataset were regular, special education, vocational education, alternative, charter, magnet, Title 1, and Title 1 schoolwide. There were not many missing values in the Title 1 column across all ten years of data pulled for this analysis. All 50 states are included in this dataset, and the U.S. territories are dropped. There were multiple years with none missing values in the columns of interest, there are a couple where there are missing values in the Title 1 schools' column.

Figure 2: Distribution of ARRA instruction funding



The missing Title I values occur for only a small number of states-year observations and reflect reporting standards not met. These specific observations will be excluded from the

analytic sample rather than discarding entire years or imputing values. The missing values for each year do not exceed more than two and it will not significantly decrease sample size.

The Title 1 summary table that is above shows the change in Title 1 school wide schools there were over the course of 2005 to 2015. The intercept is not meaningful but over time, each year that passes is associated with an increase of 60 Title 1 schools across the United States.

3.3 Expenditure per pupil

The current expenditure per pupil has no missing values and they provide insight about how each state values and funds education over time. The spending is all controlled in constant 2019 dollars. Current expenditure per pupil added to the analysis will provide evidence to which states are focusing more funds towards education and will help provide longitudinal insight about states growth over the years. Seeing how the changes in the state funding per pupil education combined with adjustments in federal funding will help contribute to a clearer analysis of government investment in human capital. Spending per pupil can potentially help reduce omitted variable bias in the regression. It is likely that states with higher per-pupil spending often have different demographic profiles, stronger labor markets, and different tax structures. This statistic will help estimate what the federal shocks from ARRA funding will look like.

3.4 Amount from ARRA Instruction Funding

The American Reinvestment and Recovery Act Fiscal Year 2010 data table has no missing data in the table. All the data in the table is adjusted and reported in constant 2009 dollars. It is important to note that Hawaii only has one school district and most of the public education is funded through state taxes. This means that there is no within-state variation or distinction between districts. Since there is a statewide school district, there are less disparities between schools because all of them are receiving funds from the same pool of funding.

The large variation in ARRA funding and the fact that it is not normally distributed further argues that there are some levels of disparities between the recipient states that drives the diversity in funding. Hypothetically using the disparities in funding, macroeconomic outcomes may aid in revealing the realistic impact that education has on economic development. This may be distinctive when compared to other states, because there is only one school district, the schools are more similar in funding capabilities, and it is more uniform across the state. The education data table, when referring to the “total spending”, does not include U.S. territories such as American Samoa, Guam, the Commonwealth of the Northern Mariana Islands, Puerto Rico, or

the U.S. Virgin Islands. The U.S. territories are not being included because most data are underreported or NA and the statistics are significant outliers compared to the fifty U.S. states.

ARRA funding can be allocated to many different federal programs in education such as teacher professional development or early kindergarten support and though all aspects are critical to student success, it is not necessarily applicable as impactful to all or most students. To emphasize the impact that education funding has on economic growth, the primary focus has been adjusted to focus on the instruction portion of the ARRA funding allocations. This allows for a more direct pipeline of association between investment in young human capital and macroeconomic outcomes.

3.5 Real Median Household Income

Like unemployment data, median household income data was originally pulled for the entire U.S. but since the cost of living, taxes and population density varies so strongly across states, it will likely be more useful to pull individual state median household income over time instead of aggregating the data for the period. Another macroeconomic indicator that would provide insight about the state of the economy and what economic growth is occurring state by state.

3.6 Unemployment

Unemployment is a defining explanatory variable of the state of the economy. It serves as a macroeconomic control variable, and it captures the business cycle as well as economic growth. Recessions and education spending often work together so it is important to include unemployment in the regression. Unemployment data was initially extracted as an aggregate for the entire U.S. but after seeing the changes in construction employment data, pulling unemployment rates for each state across periods of years may be more appropriate for the analysis.

4 Empirical Estimates

Important reminder that the term ARRA stands for the American Reinvestment and Recovery Act of 2010, an educational funding program aimed at boosting economic growth in the aftermath period of the 2008 economic crisis.

Table 2: Determinants of ARRA Instructional Funding Allocations

	Regression 1	Regression 2	Regression 3
Constant	1.68e+01*** (1.080)	1.73e+01*** (1.138)	1.74e+01*** (1.144)
Poverty Rate	4.69e-01* (0.219)	6.36e-01* (0.251)	5.93e-01* (0.268)
Unemployment Rate	-4.53e-01*** (0.076)	-4.54e-01*** (0.084)	-4.59e-01*** (0.085)
Median Income	-1.09e-04*** (0.000)	-7.34e-05*** (0.000)	-7.41e-05*** (0.000)
Title I Schools	1.11e-04 (0.000)	9.50e-05 (0.000)	9.54e-05 (0.000)
Per-Pupil Spending		-2.20e-04*** (0.000)	-2.19e-04*** (0.000)
Construction			-4.30e-03 (0.009)
R ²	0.153	0.194	0.195
Adjusted R ²	0.147	0.186	0.184

Notes: This table reports ordinary least squares (OLS) estimates of the determinants of ARRA instructional funding allocations across states. The dependent variable is the amount of ARRA instructional funding received per state. Standard errors are reported in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. The sample includes all states for which complete financial and demographic data were available.

4.1.1 Table 1: Regression 1

Regression 1 examines how state-level economic characteristics predict the percentage of ARRA instructional funding a state received. The results indicate that poverty, unemployment, and median income are all statistically significant predictors of ARRA allocations, while the number of Title I schools is not. Specifically, states with higher poverty rates received significantly more ARRA instructional funding: a one-unit increase in the poverty rate is associated with a 0.469-point increase in ARRA instruction funding ($p < 0.10$). In contrast, unemployment is negatively associated with ARRA funding: each additional point of unemployment corresponds to a 0.453-point decrease in ARRA instruction ($p < 0.01$). Median income also shows a small but statistically significant negative relationship, suggesting that higher-income states received slightly less ARRA funding. The Title I variable is effectively zero and not statistically significant, indicating that the number of Title I schools does not meaningfully predict ARRA allocations in this specification. Overall, the regression explains about 15% of the variation in ARRA instructional funding across states when taking the R squared into account.

4.1.2 Table 1: Regression 2

Regression 2 in Table 1 extends the previous specification by adding per-pupil spending as an additional control. The inclusion of spending slightly strengthens the regression, increasing the R-squared to approximately 19%, and the main patterns remain consistent. Poverty continues to be positively associated with ARRA instructional funding (0.636, $p < 0.10$), while unemployment remains a strong negative predictor (-0.454 , $p < 0.01$). Median income again shows a small negative effect. The spending variable itself is statistically significant and negative, suggesting that states with higher per-pupil expenditures received marginally less ARRA instructional funding. This aligns with the policy logic of ARRA targeting: funds were intended to support states with fewer resources and greater fiscal stress. As in Regression 1, the Title I schools remain statistically insignificant, reinforcing that the Title I schools do not meaningfully influence ARRA allocations when broader socioeconomic conditions are accounted for.

4.1.3 Table 1: Regression 3

Regression 3 shifts the focus to economic outcome as the left-hand side variable, examining whether ARRA instructional funding is associated with changes in construction employment. The results show no statistically significant relationship between ARRA instruction and construction employment change: the coefficient on ARRA instruction is small (-0.189) and not significant. Instead, the regression reveals that underlying economic conditions, in this example poverty and unemployment, are the dominant predictors of construction employment change. Poverty has a large negative effect (-9.587 , $p < 0.01$), indicating that states with higher poverty rates experienced significantly larger declines in construction employment, an expected value since construction typically signals economic growth. Unemployment also has a strong negative association (-1.077 , $p < 0.01$). Median income is not significant in this regression which is unexpected. Overall, the regression explains 22 percent of the variation in construction employment change, suggesting that ARRA instructional funding did not directly influence construction employment, and that broader economic distress was the primary driver of employment declines.

Table 3: Determinants of Changes in Construction Employment

	Regression 4	Regression 5
Constant	1.48e+01* (6.666)	1.63e+01* (7.274)
ARRA Instruction	-1.89e-01 (0.219)	-3.12e-01 (0.362)
Poverty Rate	-9.59e+00*** (1.200)	-9.56e+00*** (1.232)
Unemployment Rate	-1.08e+00** (0.398)	-1.17e+00** (0.426)
Median Income	-1.04e-04 (0.000)	-1.09e-04 (0.000)
Title I Schools		-9.26e-04 (0.002)
ARRA Instruction X Title 1		1.59e-04 (0.000)
R ²	0.224	0.226
Adjusted R ²	0.217	0.216

Notes: This table reports ordinary least squares (OLS) estimates of the determinants of changes in construction employment across states. The dependent variable is the state level change in construction employment. Standard errors are reported in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

4.2.1 Table 3: Regression 5

Regression 5 contributes to Regression 4 as well as tests whether the effect of ARRA funding on construction employment varies with the concentration of Title I schools and if Title I schools impact construction employment. The coefficient on ARRA percent funding reflects the effect of ARRA funding in a hypothetical state with zero Title I schools, while the coefficient on Title 1 represents how construction employment changes with Title I concentration when ARRA funding is zero. The key parameter is the interaction term, which indicates whether the marginal effect of ARRA funding grows stronger based on Title I school levels. A positive and significant interaction term would imply that ARRA funding is more strongly associated with construction employment growth in states with more Title I schools, suggesting heterogeneous treatment intensity across educationally disadvantaged areas.

Including an interaction term between ARRA funding and the number of Title I schools is valuable because it allows the analysis to capture meaningful heterogeneity in how stimulus dollars translated into construction employment across states with different levels of educational disadvantage. Without the interaction, the regression would impose the unrealistic assumption that ARRA funding has the same effect everywhere, regardless of underlying need, fiscal

capacity, or infrastructure conditions of individual states. The interaction term instead tests whether ARRA's impact varies systematically with Title I concentration, whether funding was more effective in states with greater school-related welfare needs, or whether its influence diminished in those contexts. By allowing the slope of ARRA funding to change with Title I levels, the regression provides a more nuanced and policy relevant understanding of how federal aid interacted with local conditions to shape macroeconomic outcomes, in this case construction employment. Unfortunately, these variables are statistically insignificant and therefore do not provide adequate information to support our hypothesis.

4.2.2 Table 4: Regression 6

Regression 6 evaluates whether ARRA funding is associated with changes in median household income. The coefficient on ARRA percent funding represents the partial association between ARRA funding and income growth, controlling for poverty and unemployment. The coefficient is positive and can be interpreted that a 1 percent increase in the percent of ARRA funding that a state receives is associated with a 7.71 dollar increase in median household income. This value is statistically insignificant, but it does provide accurate direction of coefficient to support the hypothesis. The coefficients on poverty and unemployment capture how pre-ARRA economic conditions relate to subsequent income changes. The coefficient on poverty rate is negative and statistically significant indicating that an increase in the poverty rate is associated with a 1.47 dollar decrease in median household income. Given that median household income is used to determine whether someone falls below the poverty threshold, if this value was not statistically significant, there would be cause for concern for the entire model. The unemployment rate coefficient is negative and suggests that a one unit increase in the unemployment rate is associated with a .542 dollar decrease in median household income. This regression provides a broad measure of whether ARRA funding is correlated with improvements in household economic wellbeing.

Table 4: Determinants of Changes in Median Income

	Regression 6	Regression 7	Regression 8
Constant	6.75e+02 (759.469)	6.28e+02 (1196.085)	4.44e+02 (1841.054)
ARRA Instruction	7.71e+00 (52.298)	-1.34e+02 (353.217)	-9.13e+01 (477.463)
Poverty Rate	-1.47e+03*** (305.015)	-1.46e+03*** (316.772)	-1.46e+03*** (317.105)
Unemployment Rate	-5.42e+01 (96.567)	-5.35e+01 (106.145)	-5.33e+01 (106.265)
Per-Pupil Spending		4.10e-03 (0.056)	1.75e-02 (0.116)
Title I Schools		-2.24e-02 (0.210)	-1.56e-02 (0.216)
ARRA Instruction X Log (Title 1)		2.11e+01 (54.479)	1.93e+01 (56.234)
ARRA Instruction X Per Pupil Spending			-2.41e-03 (0.018)
R ²	0.074	0.072	0.072
Adjusted R ²	0.068	0.060	0.058

Notes: This table reports ordinary least squares (OLS) estimates of the determinants of changes in median household income across states. The dependent variable is the state-level change in median household income over the study period. Standard errors are reported in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. The sample includes all states with complete demographic, financial, and income data.

4.2.3 Table 4: Regression 7

Regression 7 adds on to Regression 6 while adding the context of per pupil spending, Title 1 schools, and an interaction term of ARRA spending and a logarithmic variable on Title 1 schools. When more educational outcomes were included, the ARRA variable flipped from being positive and small to being large and negative. The value on ARRA spending in Regression 6 states that a 1 percent increase in ARRA spending is associated with a -134 dollar decrease in median household income. This is strange given that it has completely flipped from the previous regression's results, however it is still statistically insignificant. Once again poverty is statistically significant and did not change much from the previous regression. The coefficient on unemployment rate can be interpreted as a 1 percent increase in the unemployment rate is associated with a .535 dollar decrease in median household income. Per pupil spending coefficient can be interpreted as a 1 dollar increase in the per-pupil spending is associated with a 1 dollar increase in the median income. This can be rationalized because it shows that increases

in income are associated with increases in per pupil spending, theoretically because of higher income states spending more money on education per pupil because those states collect more money from taxes. Title 1 schools have a negative coefficient, explaining that a 1 percent increase in Title 1 schools is associated with a 0.0211 decrease in median income. This is rational because Title 1 schools typically are dictated by the amount of poverty concentrated in one area. The interaction term coefficient on ARRA and logarithmic Title 1 schools states that the relationship between ARRA funding and median income becomes stronger as Title 1 increases by 1 percent. This shows that ARRA funding matters more in terms of median income when Title 1 schools increase by a percentage point.

4.3.1 Table 4: Regression 8

Regression 8 adds onto Regression 7 by the addition of an interaction term between ARRA funding and per-pupil spending. Most of the other coefficients remain the same in direction even if there is a change in magnitude. Poverty rate is still the only statistically significant variable in the entire regression. The interaction term negative coefficient on ARRA and per pupil spending states that the relationships between ARRA funding and spending per pupil becomes weaker as spending increases by 1 dollar. This shows that ARRA funding matters less in terms of median income when spending increases by a dollar. This supports the hypothesis that funding would be stronger when there is lower income.

5 Discussion

Many underlying themes are appearing in this analysis but there are certain ones that stick out. Statistical significance was not a major factor across the board in regressions but is necessary to consider when considering the interpretations of variables that contradict the hypothesized values.

In Table 1, Regression 1, there is a negative statistically significant value between unemployment rate and ARRA instructional funding allocation. This signals that an increase in the unemployment rate, typically associated with a downward trend in economic growth, is associated with a decrease in ARRA funding. An unexpected result such as this signals that unemployment rate was not a contributory factor considered when the funds were being allocated by the federal government. An argument for this being the case is that coming off of a severe recession, such as the recession of 2008, unemployment being a very volatile economic

indicator, it makes sense that this was discarded in terms of consideration in the allocation conversation.

In Table 1, Regression 2, spending per pupil is a statistically significant variable. This variable is crucial to include but not without the caveats. Spending does not measure the strength of education and institutional robustness, but it does signal which states may value education higher than others. Since the federal government did take this into account when making decisions on ARRA allocations, one could argue it borderline negatively impacts the states who prioritize and pour more funding into education. A caveat like this could negatively impact the state's future decisions knowing that if more money is invested in education from the state, less money will be allocated to them when stimulus packages like the American Recovery and Reinvestment Act comes out.

In Table 1, Regression 3, construction employment was not a significant predictor of ARRA funding. This was an unexpected result given that the other economic outcomes were statistically significant. In terms of allocations, this could be explained by the fact the similar caveat of the unemployment rate having a negative effect on ARRA in all the regressions of the table. Unemployment is a rather volatile variable and especially with a significant economic downturn such as a deep recession, unemployment and employment no matter the sector chosen would be less likely considered. Currently there is no explanation as to why unemployment overall is statistically significant at the one percent level and why construction employment is not statistically significant at all.

In Table 2, Regression 3, median income is not statistically significant. This is not expected and it is a negative variable value which is odd because one would assume that an increase in income would be associated with an increase in employment. However, contrary to the biotechnology sector which requires a higher degree, taking note that higher education is associated with higher median income, in this case having higher median income is associated with lower employment outcomes makes sense. Some people who work in construction have high incomes however this is not the case for most of the manual laborers who are in the construction sector. In the end, median income is not significant, however, it is interesting to interpret the values, nonetheless.

In Table 2, Regression 4, the coefficient for Title 1 schools is negative and statistically insignificant. Signaling that an increase in Title 1 schools is associated with a decrease in

construction employment. This is an expected result especially based on our results in Table 1. There is another term however, an interaction term between ARRA funding and Title 1 schools. This coefficient is positive, indicating that the impact of ARRA funding on construction employment is stronger when there are more Title 1 schools.

In Table 3, Regression 5, the coefficient on ARRA spending is positive. This coefficient is difficult to interpret because it could go one of two ways. The value is positive because an increase in ARRA funding helps build the economy, having a positive effect on median income supports this theory. On the other hand, ARRA funding was given to states that were poorer at the time and was a single event, so at the time of the study, it is expected that the higher the ARRA funding is, the lower grade, economically, that state was. It is also possible that within the scope of the data, it does not allow for interpretation into the long-term future since the data stops at 2015.

In Table 3, Regression 6, the coefficient on ARRA funding flipped to negative, however, remained statistically insignificant. This can be explained by the fact that the ARRA funding did not change over time and was a one-time occurrence. The increase in ARRA funding at the one time, helped indicate what the economic needs were of one state at one time. This could make sense in terms that the ARRA funding was allocated based on poorer economic situations so one could justify a negative and strong coefficient on ARRA funding, especially after the results in Table 1.

Compared to the background literature, it seems as though there are many variables missing and can be valuable when using this data to create action plans and considerations for other funding opportunities in the future. Creating a threshold variable and taking school quality into account, like Trabelsi, would be a beneficial consideration moving forward since it seems that these are better indicators of economic performance.

Like Clarke et al. with the biotechnology sector, using the construction employment variable provided some insight but since construction workers' median income is lower than the median income, it turned out to not be a helpful predictor. Picking an employment sector where the median income of that sector is closer to the overall median income for that state would likely be more beneficial when predicting economic outcomes and reasoning about federal education allocations moving forward.

6 Conclusion

Many considerations were taken in all the analyses' discussed throughout this literature. The background information provided rational concepts and arguments as to why governments tend to lean towards a certain result. The empirical analysis examined how federal funds were allocated across states and whether this variation in funding is associated with measurable and statistically significant economic outcomes in the years following the Great Recession. The results showed that preexisting conditions, such as poverty, unemployment and median income were some of the strongest predictors of ARRA funding levels, whereas education indicators like Title 1 schools were less significant. This adds to the literature that quality and quantity should be taken into consideration as well when discussing human capital impacting economic growth. The growth variables are also vital pieces of the puzzle, where the right piece has yet to be found.

Throughout this paper and previous empirical papers discussing the same concept, many different methods were utilized. After this discussion, it remains a mystery which economic outcome best measures the power and strength that education spending has on the economic wellbeing of a country. This analysis did make some previous assumptions more robust, especially the argument that K-12 schools have a lesser impact on economic outcomes compared to higher education. In the US, it seems that this is the case given that it is broadly accepted that high school diplomas and fully funded public education levels are no longer considered to be strong enough to guarantee economic wellbeing. In the future, it will likely be beneficial to consider more variables and potentially apply more case study analyses to get a better grasp on true economic outcomes of education spending in the United States. The level of complexity in this topic has yet to be simplified but continued studies hopefully will reveal more of what truly makes an impact.

7 References

- Bachmann, M. (n.d.). *Charting the progress of education reform: An evaluation of the recovery act's role*. IES. <https://ies.ed.gov/use-work/awards/charting-progress-education-reform-evaluation-recovery-acts-role#:~:text=Key%20outcomes,and%20varied%20by%20reform%20area>
- Clarke, W., Jones, A. T., Lacy, B. (2015). Education spending and workforce quality as determinants of economic growth. *The Journal of Rural and Community Development*, 10(4), 24-35.
- Deskins, J., Hill, B., & Ullrich, L. (2009). Education Spending and State Economic Growth: Are All Dollars Created Equal? *Economic Development Quarterly*, 24(1), 45-59. <https://doi.org/10.1177/0891242409347370> (Original work published 2010)
- Hanushek, E.A., Woessmann, L. Do better schools lead to more growth? Cognitive skills, economic outcomes, and causation. *J Econ Growth* 17, 267–321 (2012). <https://doi.org/10.1007/s10887-012-9081-x>
- Pirim, Zafer; Owings, William A.; and Kaplan, Leslie S., "The Long-Term Impact of Educational and Health Spending on Unemployment Rates" (2014). Educational Foundations & Leadership Faculty Publications. 16. https://digitalcommons.odu.edu/cfl_fac_pubs/16
- Trabelsi, S. (2018). Public Education Spending and Economic Growth: The Governance Threshold effect. *Journal of Economic Development*, 43(1).
- U.S. Bureau of Labor Statistics, All Employees, Construction [USCONS], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/USCONS>, March 7, 2026.
- U.S. Department of Education, National Center for Education Statistics, Common Core of Data (CCD), "National Public Education Financial Survey (NPEFS)," fiscal year 2010, provisional Version 1a.